JavaScript

Conditional Statements

01 Conditional Statements

02 Comparison Operators

03 IF, ELSE IF, ELSE Statements

04 Switch Statement

Conditional Statements

- Are statements that executes actions depending on different conditions.

- They are used with Comparison Operators and Logical Operators

- In JavaScript we have the following conditional statements:
  - Use **if** to specify a block of code to be executed, if a specified condition is true
  - Use **else** to specify a block of code to be executed, if the same condition is false
  - Use **else if** to specify a new condition to test, if the first condition is false
  - Use **switch** to specify many alternative blocks of code to be executed

Comparison Operators

- Are Operators used to compare 2 values.

- Always outputs true or false

Operator	Description
>	Greater Than
<	Less Than
>=	Greater Than or Equal
<=	Less Than or Equal

Comparison Operators

Operator	Description
==	Equal
===	Equal Value and Type
!=	Not Equal
!==	Not Equal Value and Type

Comparison Operators

JS Tutorial # 4 > JS script.js > ...

```
let x = 5;
console.log (x == "5")
//Output : true

console.log (x === "5")
//Output : false


console.log (x != "5")
//Output : false

console.log (x !== "5")
//Output : true
```

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```
true                    script.js:2
false                   script.js:5
false                   script.js:9
true                    script.js:12
```

Comparison Operators

```
15
16  console.log(10 > 10);
17  //Output : false
18
19  console.log(10 < 10);
20  //Output : false
21
22
23  console.log(10 >= 10);
24  //Output : true
25
26  console.log(10 <= 10);
27  //Output : true
```

```
false                                   script.js:16
false                                   script.js:19
true                                    script.js:23
true                                    script.js:26
>
```

If statement

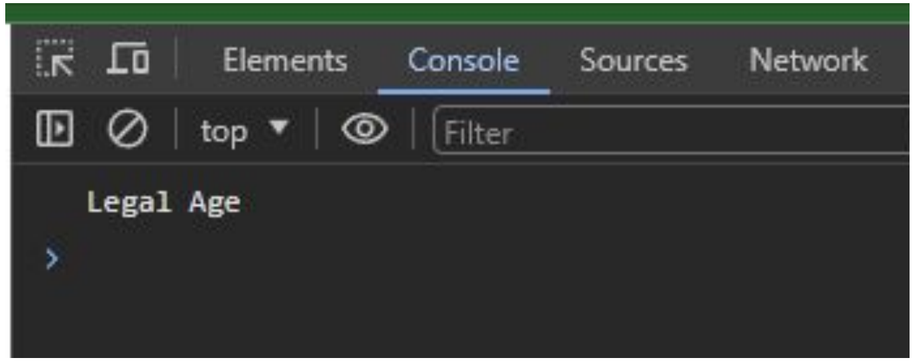
- Use the if statement to specify a block of JavaScript code to be executed if a condition is true.

- Syntax

```
if (condition){
    // block of code to be executed if the condition is true
}
```

```
//Print "Legal Age" if age is greater than or equal to 18
let age = 18;

if ( age >= 18 ){
    console.log("Legal Age")
}
```



else statement

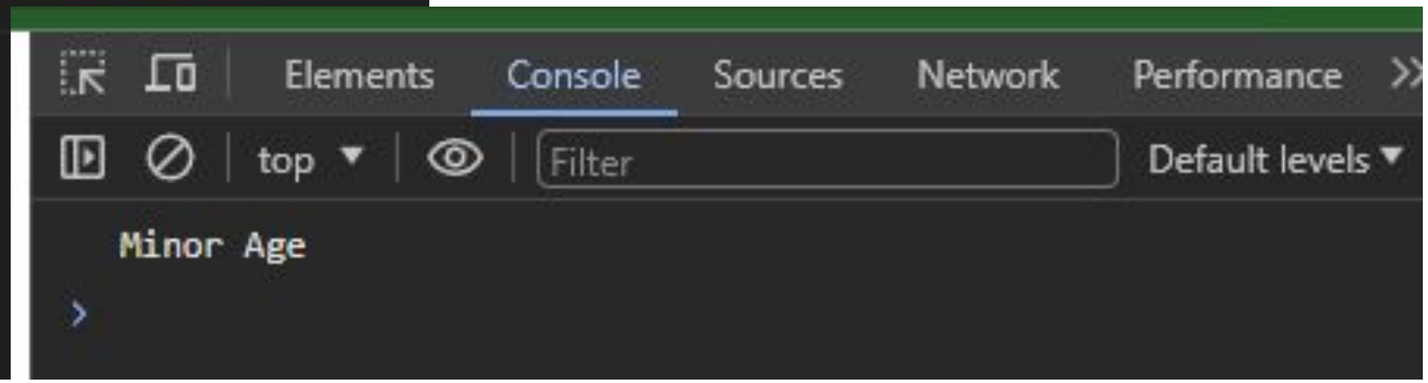
- Use the else statement to specify a block of code to be executed if the condition is false.

- Syntax

```
if (condition){
    // block of code to be executed if the condition is true
}else{
    // block of code to be executed if the condition is false
```

```
//Print "Legal Age" if age is greater than or equal to 18 or Print "Minor Age"
let age = 17;

if ( age >= 18 ){
    console.log("Legal Age")
}else{
    console.log("Minor Age")
}
```



else if statement

- Use the else statement to specify a block of code to be executed if the condition is false.

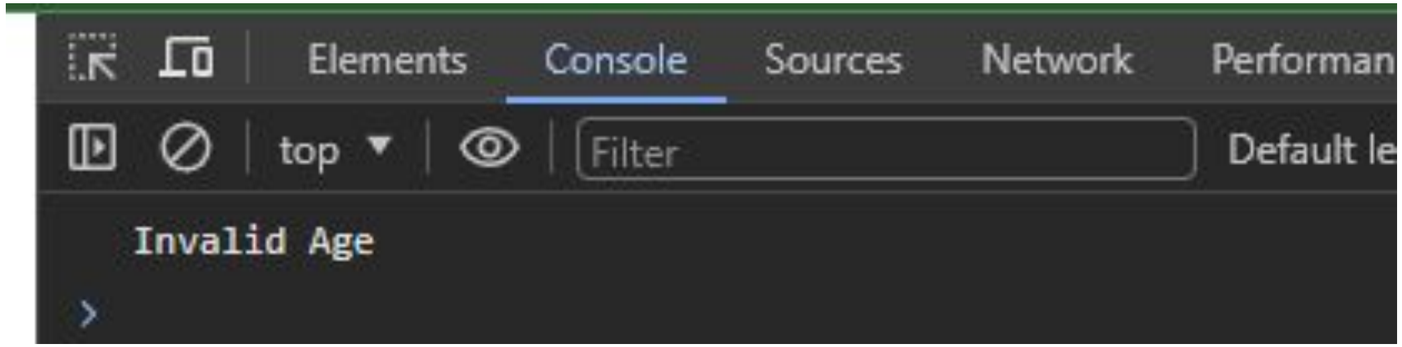
- Syntax

```
if (condition){
    // block of code to be executed if the condition is true
}else if (condition){
    // block of code to be executed if the condition is true
else{
    // block of code to be executed if the condition is false
}
```

else if statement

```
// Print "Invalid Age" if age is less than or equal to 0
// Else if Print "Legal Age" if age is greater than or equal to 18
// Else Print "Minor Age" if age is less than 18
let age = 0;

if(age <=0){
    console.log("Invalid Age")
}else if ( age >= 18 ){
    console.log("Legal Age")
}else{
    console.log("Minor Age")
}
```



Switch Statement

- Use the switch statement to select one of many code blocks to be executed.

- This is how it works:
  - The switch expression is evaluated once.
  - The value of the expression is compared with the values of each case.
  - If there is a match, the associated block of code is executed.
  - If there is no match, the default code block is executed.

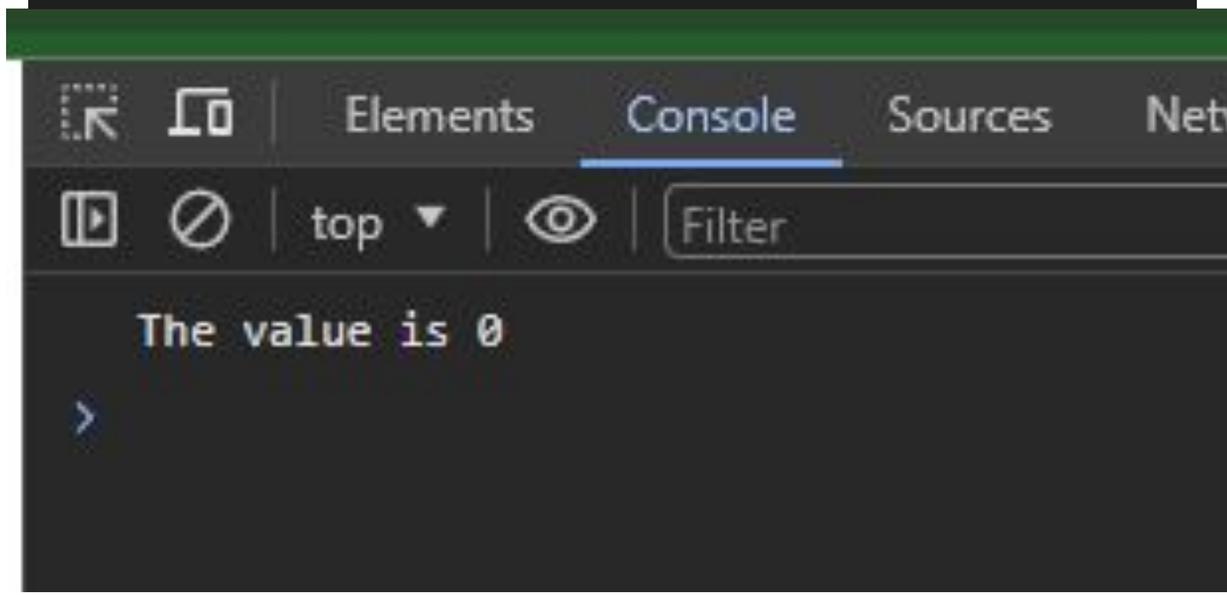
Switch Statement

- Syntax

```
switch(expression) {
  case x:
    // code block
    break;
  case y:
    // code block
    break;
  default:
    // code block
}
```

```
let number = 0;
let output  = "";
switch(number){
    case 0 :
        output = "0";
        break;
    case 1  :
        output = "1"
        break;
    default:
        output = "Not found."
}

console.log(`The value is ${output}`)
```



JavaScript

Thank you

Reference :

- https://www.youtube.com/watch?v=V01uCwftQ5o&list=PLVnJhHoKgEmp5SRFPKsy61BeWozI3mwcV&index=1
- https://www.w3schools.com/js/js_if_else.asp